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(54) Title: A DEVICE FOR MEASURING REAL TIME OBSTETRIC BLOOD LOSS

(57) Abstract: A device for measuring real time obstetric blood loss comprising a bowl connected to a collection bag characterized by the fact that the bowl (1) is flexible, has a broad mouth (2) with a thick rim (3), a narrow tapering base (4) connected by a tube (5) to two outlet ports (6,7), one outlet port (6) leading to a calibrated collection bag (9) and the other outlet port (7) connected to a suction device (10) having a calibrated collection unit (11).

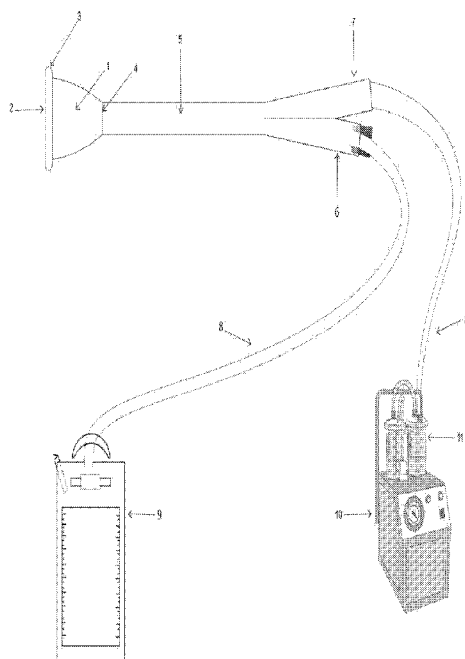


FIGURE 2



TITLE: A DEVICE FOR MEASURING REAL TIME OBSTETRIC BLOOD LOSS**5 FIELD OF INVENTION:**

The present invention relates to a device for measuring real time obstetric blood loss. In particular, the device of the present invention measures real time obstetric blood loss thereby preventing maternal mortality and morbidity.

10 PRIOR ART:

Obstetric blood loss is one of the major causes of maternal mortality and morbidity. Of the different kinds of obstetric blood loss, Postpartum Hemorrhage (PPH) is said to be responsible for 25-55% of maternal deaths. Globally, it has been found that one mother loses her life every 4 minutes due to PPH. PPH is unpredictable and hence
15 every parturient woman is at risk of suffering from PPH.

To curb maternal mortality, early diagnosis even before appearance of clinical signs and symptoms is necessary, so that the “Golden Hour” i.e. the first sixty minutes after delivery, is not lost. With the passage of time, the complexity of any given case is likely to increase because continuous bleeding, not appropriately and adequately
20 controlled on a timely basis, invariably leads to coagulopathy which will ultimately lead to maternal mortality. Hence, medical teams at hospitals need to be equipped to predict, prepare and handle such blood loss at the earliest available opportunity. The accurate estimation of blood loss is the most important and key factor for treating PPH, even before clinical signs of PPH appear, as the actual blood loss is often
25 underestimated in the present clinical practice. Usually, obstetricians, gynecologists and anesthesiologists receive distress calls for PPH at odd hours, which causes delay in diagnosis and requires marathon efforts by the medical team to save the life of the patient.

Apart from blood loss due to PPH, blood loss during pregnancy, or in the case of still birth or abortion, the blood loss either immediately after the procedure or days after the procedure have also been found to be responsible for maternal mortality and morbidity. The real time estimation of blood loss is therefore of utmost importance.

5 Currently, the widely used methods to estimate obstetric blood loss are:

- Visual estimation
- Use of Brass V Drapes
- Use of Kelly's Pad

Visual estimation is considered to be a highly inaccurate method. Studies have found
10 visual estimation of blood loss to be insensitive in diagnosing obstetric blood loss and has also been found to underestimate postpartum blood loss by up to 50%, which may result in maternal mortality or severe maternal morbidity.

Use of Brass V Drapes and Kelly's pad do not give real time and accurate measurement of blood loss; they are cumbersome to use; there is scope for spillage;
15 they can be uncomfortable for patients; maintenance of hygienic conditions at all times becomes difficult; and they are not reusable, thereby increasing bio-medical waste.

CN202724124U discloses a utility model to provide postpartum nursing pad dedicated to collecting and measuring postpartum hemorrhage. It includes a
20 waterproof pad and elastic band, characterized in that a funnel-like drainage groove is provided on the waterproof pad and the lower end of the funnel-like drainage groove is provided with a scale drainage bag. Such a pad though simple in structure, is inconvenient to use for the parturient woman as it has to be placed below the buttocks, which would need lifting of the woman from the hips; it is a use and throw
25 pad and hence adds to bio-medical waste; accurate blood loss will not be measured if the pad is not placed in the correct position and all the blood is not likely to be collected in the drainage bag; it can be cumbersome to use and maintain hygienic conditions at all times; and it is not easy to shift the pad if the woman is being shifted from the delivery table to a bed.

There is therefore a need for a device which does not contain any of the aforesaid disadvantages of the prior art and which is able to accurately estimate/measure real time obstetric blood loss.

5 **OBJECT OF THE INVENTION:**

The object of the present invention is to provide a device for measuring real time obstetric blood loss, especially in the Golden Hour and prevent late diagnosis.

Another object of the present invention is to provide a device for measuring real time obstetric blood loss, which is reusable upon sterilization, thereby ensuring zero bio-
10 medical wastage.

Yet another object of the present invention is to provide a device that does not require highly trained staff to use or to monitor its effective functioning.

Yet another object of the present invention is to provide a device that also helps to easily drain blood clots clogging the device, thereby ensuring steady and
15 uninterrupted flow of blood through the device into the collection bag.

Yet another object of the present invention is to provide a device which removes blood clots from the inner uterine wall, thereby allowing the uterus/uterine wall to recover speedily and get back to its normal size.

Yet another object of the present invention is to provide a device in which the
20 negative pressure used for removing blood clots can be adjusted depending on the need of the patient.

Yet another object of the present invention is to prevent spillage and minimize any possibility of infection, thereby not hampering the health of the patient, and minimizing exposure to the doctors and staff treating the patient as well.

25 Yet another object of the present invention is to provide a device which has a simple design and assembly.

Yet another object of the present invention is to provide a device that is practical, convenient, comfortable and easy to use, hygienic, economical and eco-friendly.

Yet another object of the present invention is to provide a method for measuring real time obstetric blood loss.

5

SUMMARY OF THE INVENTION:

The present invention relates to a device for measuring real time obstetric blood loss comprising a bowl connected to a collection bag characterized by the fact that:

10 the bowl is flexible, has a broad mouth with a thick rim, a narrow tapering
 base connected by a tube to two outlet ports;

 one outlet port leading to a calibrated collection bag;

 and the other outlet port connected to a suction device having a calibrated
 collection unit.

15 Typically, a device for measuring real time obstetric blood loss, wherein the device
 is made of a polymer.

Typically, a device for measuring real time obstetric blood loss, wherein the device
is made of silicone.

Typically, a device for measuring real time obstetric blood loss wherein the outlet
port(s) are connected to the collection bag/ unit by means of a flexible tube (s).

20 Typically, a device for measuring real time obstetric blood loss wherein either one
 of the outlet port (s) is plugged.

Typically, a device for measuring real time obstetric blood loss wherein both the
outlet ports are plugged.

25 Typically, a device for measuring real time obstetric blood loss wherein the plug is a
 valve.

BRIEF DESCRIPTION OF THE DRAWINGS:

The following drawings are illustrative of particular examples for enabling the working of the device of the present invention. They are descriptive of the invention and are not intended to limit the scope of the invention. The drawings are not to scale
5 (unless so stated) and are intended for use in conjunction with the explanations in the following detailed description.

In the accompanying drawings:

Figure 1 shows the front view of the device for measuring real time obstetric blood
10 loss.

Figure 2 shows the front view of the said device with one outlet port attached to a calibrated collection bag by a flexible tube and the other outlet port attached to a suction device by a flexible tube.

Figure 3 shows the said device fastened/lodged on the inner wall of the birth
15 canal/vagina.

Figure 4 shows the front view of the said device having one outlet port attached to a suction device and the other outlet port which is plugged.

It will be understood that certain elements may be intentionally omitted from a particular view where they are believed to be illustrated for better advantage in the
20 other view.

The following are the details of the main components of the device in accordance with the present invention which are used with reference to Figs. 1 to 4.

1 = Bowl

25 2 = Mouth of the bowl

3 = Rim

4 = Base of the bowl

5 = Tube

6, 7 = Outlet Ports

8 = Flexible Tube

9 = Calibrated Collection bag

5 10 = Suction device

11 = Calibrated Collection unit

12 = Birth canal/vagina

13 = Plug

10 Persons skilled in the art will appreciate that elements in the figures are illustrated for simplicity and clarity and may have not been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various exemplary embodiments of the present disclosure. Throughout the drawings, it should be noted that like
15 reference numbers are used to depict the same or similar elements, features, and structures. However, the drawings and/or illustrations in no way limit the scope of the invention. It is understood that modifications and/or alterations of the drawings/illustrations of this invention other than those specifically set forth herein may be achieved by those skilled in the art and that such modifications and alterations
20 are to be considered as falling within the overall scope of this invention.

DESCRIPTION OF THE INVENTION:

The present invention will now be described with reference to the accompanying drawings wherein the same numerals are used to denote the same part.

25 The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the invention. The description given herein is only

representative of the illustrative embodiments and examples. Accordingly, it should be apparent to those skilled in the art that the following description of exemplary embodiments of the present invention are provided for illustration purpose only and not for the purpose of limiting the invention as defined by the appended claims and
5 their equivalents.

Figure 1 shows the front view of the device for measuring real time obstetric blood loss. The said device comprises of a bowl (1); a tube (5); and two outlet ports (6,7).

The bowl (1) of the device, is wide at the top and tapers towards its base. The bowl (1) comprises of a mouth (2) having a rim (3) and a narrow tapering base (4). The
10 diameter of the mouth (2) of the bowl (1) is greater than the diameter of the base (4) of the said bowl. The rim (3) on the mouth (2) of the bowl (1) is thicker than the rest of the wall of the bowl (1). The bowl (1) of the device is to be lodged/fastened inside the birth canal/vagina (not shown). In a preferable embodiment of the present invention, the said device is made of a polymer. In another preferable embodiment
15 of the present invention, the said device is made of silicone. This enables the said device to be flexible and be easily inserted inside the birth canal/vagina by partially folding the bowl (1) during insertion and thereafter releasing the folds once inside the birth canal/vagina so that it can fasten/lodge itself on the inner wall of the birth canal/vagina. The thickness of the rim (3) on the mouth (2) enables the bowl (1) to
20 firmly fasten/lodge itself onto the inner wall of the birth canal/vagina and form a water tight contact with the said inner wall of the birth canal/vagina to prevent any leak/spillage of blood flowing out of the uterus (not shown).

In a preferable embodiment of the present invention, the bowl (1) is conical in shape. In yet another preferable embodiment of the present invention, the bowl (1) is in the
25 shape of a cup. In yet another preferable embodiment of the present invention, the bowl is a menstrual cup. When the bowl (1) of the device is fastened/lodged on the inner wall of the birth canal/vagina, the blood that flows out of the uterus (not shown) directly enters the bowl (1) seamlessly.

From the bowl (1), the blood from the uterus flows through the tube (5). The base of the tube (5) is forked and ends into two outlet ports (6,7). The tube (5) connects the bowl (1) to the outlet ports (6,7).

In a preferable embodiment of the present invention, the base of the tube (5) has a
5 single outlet port.

Figure 2 shows the front view of the said device with one outlet port (6) attached to a collection bag (9) by a flexible tube (8) and the other outlet port attached to a suction device (10) also by a flexible tube (8). The blood flows from the tube (5) through the outlet port (6) by gravity and gets collected in the collection bag (9). The collection
10 bag (9) has measurements printed on its outer surface which enables one to accurately measure the real time obstetric blood loss. The other outlet port (7) is attached to a suction device (11).

The blood that flows out from the uterus is not always fluid and often contains blood clots. These blood clots clog/block the base (4) of the bowl (1) and prevent the blood
15 to flow out into the tube (5). Thus, the suction device (10) attached to outlet port (7) is periodically used to remove/unclog/unblock the base (4) of the bowl (1). The suction device (10), when in use, creates a negative pressure inside the birth canal/vagina (not shown) which facilitates the blood clots to be sucked out from the base (4) of the bowl (1) and out through outlet port (7) into the collection unit (11)
20 forming part of the suction device (10), which also has measurements printed on its surface. The outlet port (7) is an essential component of the said device as it prevents blocking of the base (4) of the bowl (1) by blood clots. The suction device (10) not only creates a negative pressure inside the tube (5) but also inside the birth canal/vagina and the uterus which helps to remove blood clots stuck to the inner wall
25 of the birth canal/vagina or formed on the inner uterine wall, which would otherwise have to be removed physically by the doctor by use of fingers or hand as the case may be. This helps the uterine wall to recover speedily and go back to its original size. It further reduces scope of any infection or unnecessary exposure to the doctor and patient.

In a preferable embodiment of the present invention, when the device is not in use or when the patient is being shifted from the operation table to the bed, the outlet ports (6,7) are closed by means of a plug (13). This prevents spillage of blood. In a preferable embodiment of the present invention, the plug (13) is a valve.

5 Figure 3 shows the said device fastened/lodged on the inner wall of the birth canal/vagina (12) of a woman. In case of vaginal delivery, the bowl (1) of the said device is inserted into the birth canal/vagina (12) immediately after delivery of the child. In case of cesarean delivery, the bowl (1) may be fastened/lodged onto the inner wall of the birth canal/vagina (12) before, during or after surgery. Once
10 fastened/lodged inside the birth canal/vagina (12), the blood flows freely through the bowl and into the collection bag (9) and/or the collection unit (11) as the case may be.

The said device can also be used in the manner as described above for measuring real time obstetric blood loss during pregnancy or in case of still birth or abortion, during
15 or after the procedure.

Figure 4 shows a device which has one outlet port (7) attached to a suction device (10) having a collection unit (11) and wherein the other outlet port (6) is plugged by a plug (13). The suction device (10) is used periodically to facilitate suction of blood clots clogging/blocking the base (4) of the bowl (1) and those formed inside the
20 uterine wall. The blood flows through the bowl (1), into the tube (5) and through the outlet port (7) into the collection unit (11) attached to the suction device (10).

In a preferable embodiment of the present invention, the said device is used for accurately measuring real time obstetric blood loss through vagina.

25 **ADVANTAGES**

1. The said device is user friendly and eco-friendly;
2. The said device can be used for measuring any type of real time obstetric blood loss through vagina;

3. The said device is comfortable to use for patients; it does not create any wetness caused by leak/spillage of blood from the vagina; it is not painful; it is hygienic and minimizes handling by doctors/nurses;

4. The said device does not require highly trained staff to handle and monitor;

5 5. The said device is safe to use and is effective in real time measurement of obstetric blood loss thereby allowing early diagnosis in the Golden Hour;

6. The said device reduces the use of sanitary pads;

7. The said device addresses the issue caused by blockage of the bowl due to blood clots with minimum handling.

10

Many features and advantages of the invention are apparent from the detailed specification, and thus, it is intended by the appendant claims to cover all such features and advantages of the invention which fall within the true spirit and scope of the invention. Further, since numerous modifications and variations will readily occur to those skilled in the art, it is not desired to limit the invention to the exact construction and operation illustrated and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the invention.

20

CLAIMS

1. A device for measuring real time obstetric blood loss comprising a bowl connected to a collection bag characterized by the fact that:
 - 5 the bowl (1) is flexible, has a broad mouth (2) with a thick rim (3), a narrow tapering base (4) connected by a tube (5) to two outlet ports (6,7), one outlet port (6) leading to a calibrated collection bag (9), and the other outlet port (7) connected to a suction device (10) having a calibrated collection unit (11).
- 10 2. A device as claimed in Claim 1, wherein the device is made of a polymer.
3. A device as claimed in Claim 2, wherein the device is made of silicone.
4. A device as claimed in Claim 1, wherein the outlet port(s) (6,7) are connected
15 to the collection bag/ unit (9/ 11) by means of a flexible tube (s) (8).
5. A device as claimed in Claim 1, wherein either one of the outlet port (s) (6,7) is plugged.
- 20 6. A device as claimed in Claim 1, wherein both the outlet ports (6,7) are plugged.
7. A device as claimed in Claims 5 and 6, wherein the plug (13) is a valve.

25

AMENDED CLAIMS
received by the International Bureau on 19 May 2021 (19.05.2021)

Claims

- [Claim 1] A device for measuring real time obstetric blood loss comprising a bowl connected to a collection bag characterized by the fact that: the bowl (1) is flexible, has a broad mouth (2) with a thick rim (3), a narrow tapering base (4) connected by a tube (5) to two outlet ports (6,7),
one outlet port (6) leading to a calibrated collection bag (9),
and the other outlet port (7) connected to a suction device (10) having a calibrated collection unit (11).
- [Claim 2] The device as claimed in Claim 1, wherein the device is made of a polymer.
- [Claim 3] The device as claimed in Claim 2, wherein the device is made of silicone.
- [Claim 4] The device as claimed in Claim 1, wherein the outlet port(s) (6,7) are connected to the collection bag/ unit (9/ 11) by means of a flexible tube (s) (8).
- [Claim 5] The device as claimed in Claim 1, wherein either one of the outlet port (s) (6,7) is plugged.
- [Claim 6] The device as claimed in Claim 1, wherein both the outlet ports (6,7) are plugged.
- [Claim 7] The device as claimed in Claims 5 and 6, wherein the plug (13) is a valve.

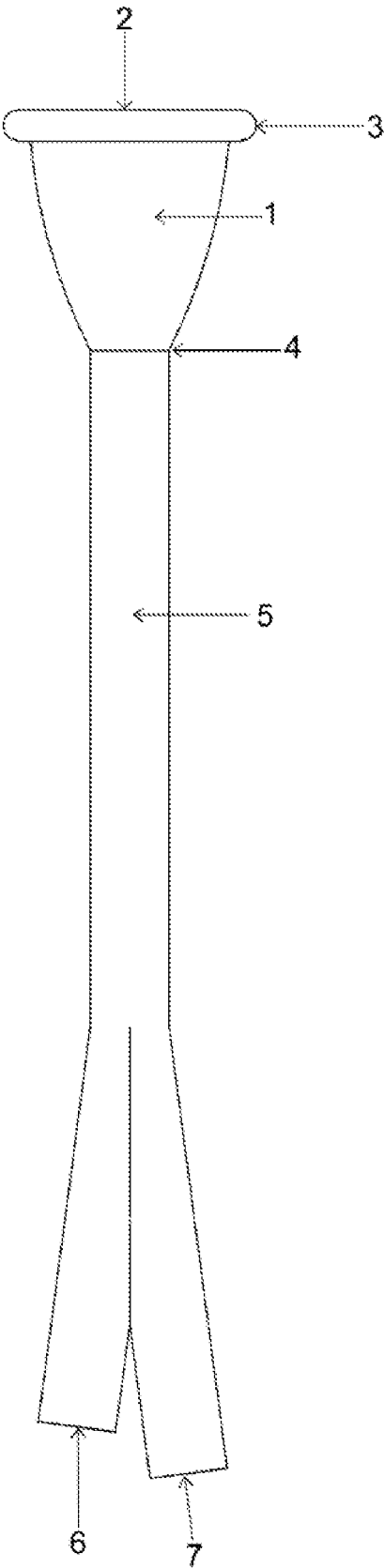


FIGURE 1

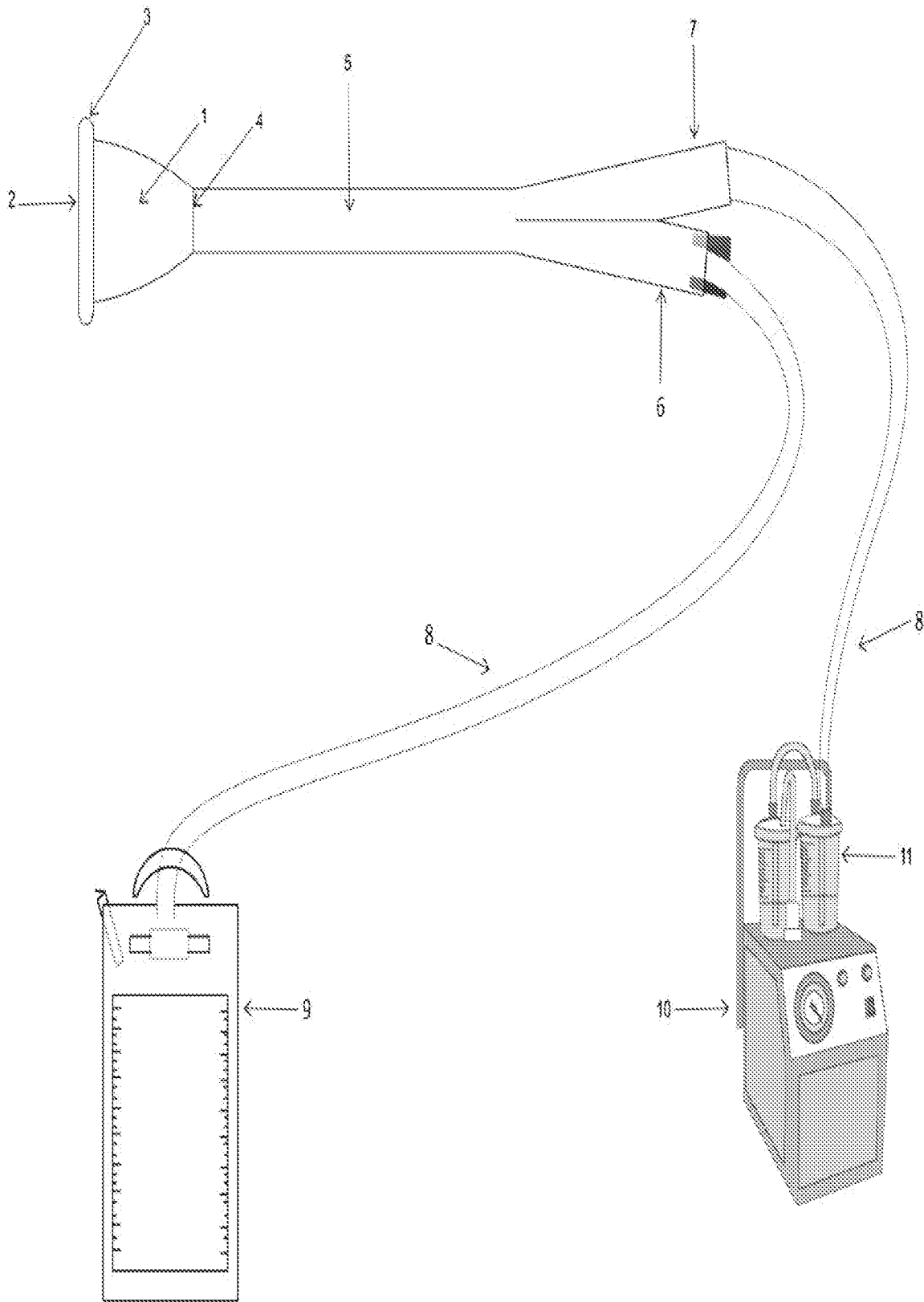


FIGURE 2

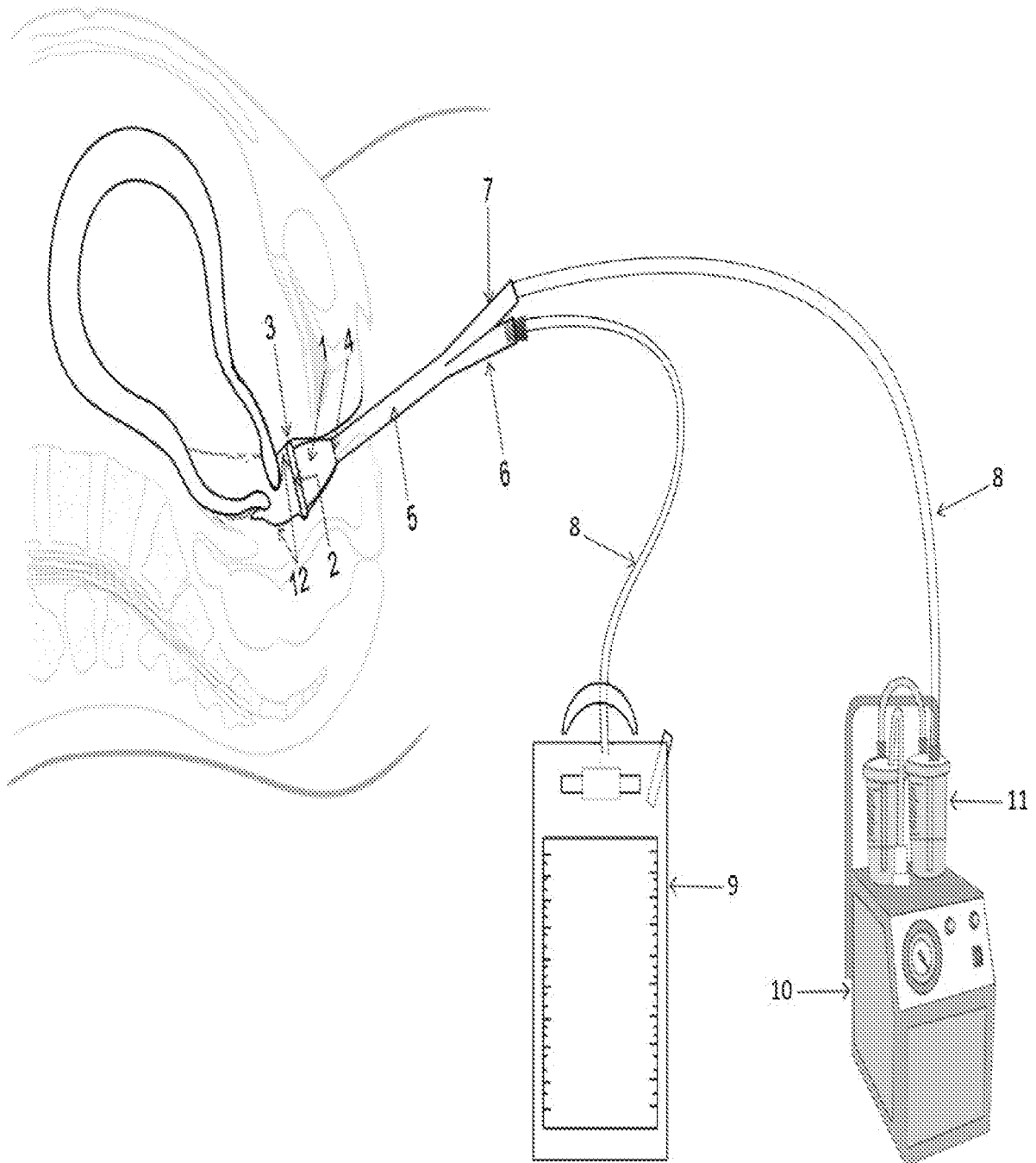


FIGURE 3

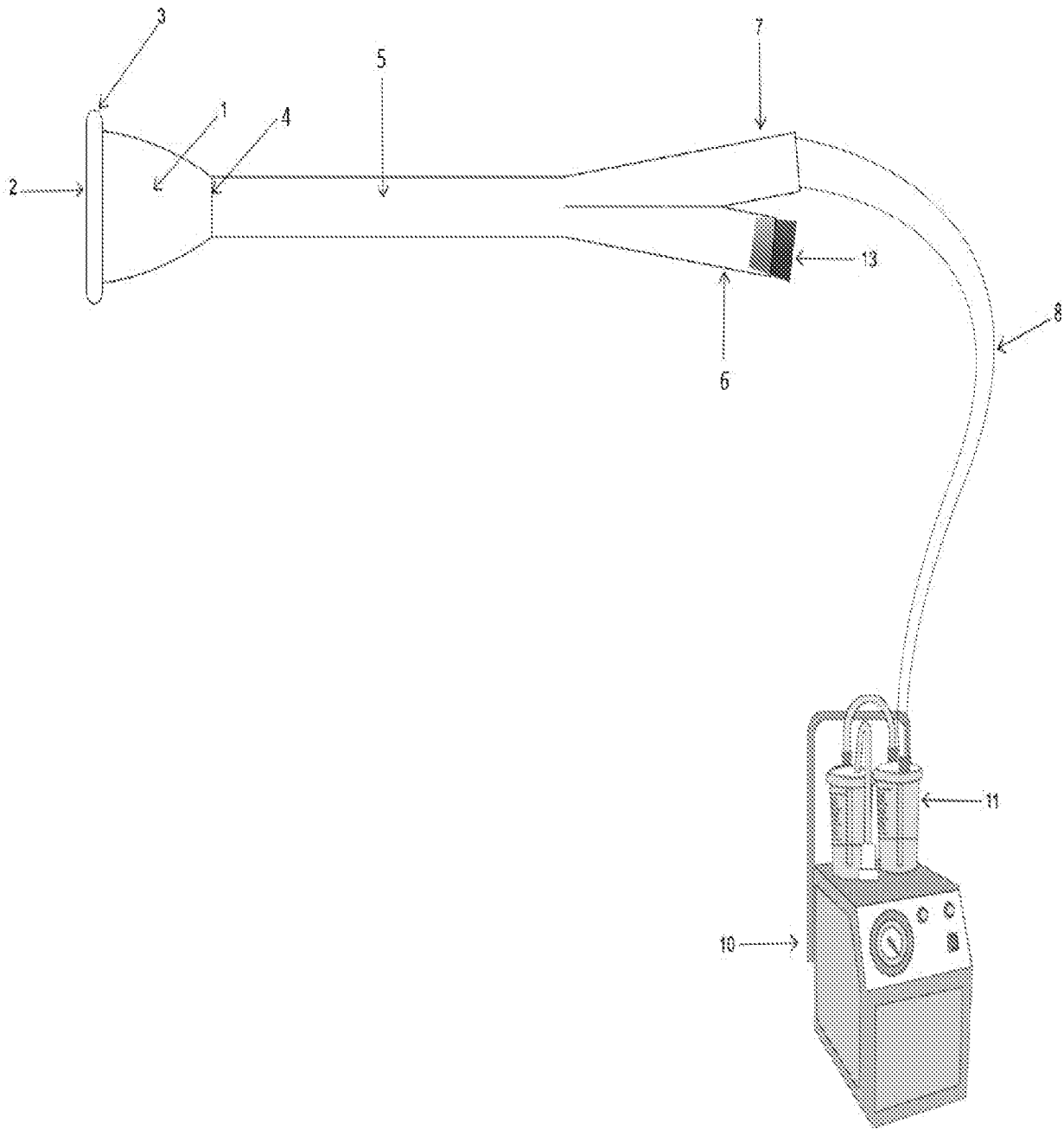


FIGURE 4

INTERNATIONAL SEARCH REPORT

International application No.
PCT/IN2021/050103

A. CLASSIFICATION OF SUBJECT MATTER
A61B5/00, A61B5/02 Version=2021.01

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

A61B

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

Databases: Total Patent One, IPO Internal Database

Keywords: obstetric blood , postpartum, collection , measurement

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	CN106390212B (West China Second Hospital of Sichuan University) 26 March 2019 (26-03-2019) Whole document	1-7
Y	CN203468737U (JIANG JING et al) 12 March 2014 (12-03-2014) Whole document	1-7
Y	CN209091500U (GUIZHOU PROVINCIAL INST OF POPULATION AND FAMILY PLANNING) 12 July 2019 (12-07-2019) Whole document	1-7

☐ Further documents are listed in the continuation of Box C. ☒ See patent family annex.

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"D" document cited by the applicant in the international application

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"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

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Date of the actual completion of the international search

23-03-2021

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23-03-2021

Name and mailing address of the ISA/

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INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/IN2021/050103

Citation	Pub.Date	Family	Pub.Date
CN 106390212 B	26-03-2019	CN 206026751 U	22-03-2017